

Chapter Six

Higher Order Linear Non-homogeneous Differential Equations with Constant Coefficients

Section-6.1

Solution of differential equation of the form $f(D)y = R(x)$, where $R(x)$ is algebraic functions

6.1.1 Higher Order Linear Non-homogeneous Differential Equations

The differential equation of the form

$$a_0 \frac{d^n y}{dx^n} + a_1 \frac{d^{n-1} y}{dx^{n-1}} + \dots + a_{n-1} \frac{dy}{dx} + a_n y = R(x) \quad \dots (1)$$

may be written as $(a_0 D^n + a_1 D^{n-1} + \dots + a_{n-1} D + a_n)y = R(x); D \equiv \frac{d}{dx} \quad \dots (2)$

where the coefficients $a_0, a_1, \dots, a_n$ are constants and function $R(x)$ are to be independent of the variable y , is called n -th order non-homogeneous linear differential equation.

6.1.2 Complementary Solution or Complementary Function (C.F.)

Let us consider the differential equations

$$(a_0 D^n + a_1 D^{n-1} + \dots + a_{n-1} D + a_n)y = R(x) \quad \dots (1)$$

and $[(a_0 D^n + a_1 D^{n-1} + \dots + a_{n-1} D + a_n)y = 0 \quad \dots (2)$

Then the general solution of the equation (2) is called the **complementary solution or complementary function** of the equation (1). Complementary solution is generally denoted by y_c .

6.1.3 Particular Solution or Particular Integral (P.I.)

Let us consider the differential equation

$$(a_0 D^n + a_1 D^{n-1} + \dots + a_{n-1} D + a_n)y = R(x) \quad \dots (1)$$

or, $f(D)y = R(x)$

Then the solution $\frac{1}{f(D)}R(x)$, which not containing any arbitrary constants, is called the **particular solution or particular integral (P.I.)** of the equation (1). Particular solution is generally denoted by y_p .

6.1.4 General Solution of Non-Homogeneous Linear Differential Equations of order n

Let us consider the differential equation

$$(a_0 D^n + a_1 D^{n-1} + \dots + a_{n-1} D + a_n)y = R(x) \quad \dots (1)$$

Also let, $y = y_c$ be the complementary solution and $y = y_p$ be the particular solution of the differential equation (1), then the general solution of the differential equation (1) is $y = y_c + y_p$

Theorem 6.1.1: If $y = y_c$ be the complementary solution and $y = y_p$ be the particular solution of the differential equation

$$(a_0 D^n + a_1 D^{n-1} + \dots + a_{n-1} D + a_n)y = R(x) \quad \dots (1)$$

then $y = y_c + y_p$ is a general solution of the differential equation (1).

6.1.5 Meaning of the Symbol $\frac{1}{f(D)}$ or Inverse Operator

$\frac{1}{f(D)}R(x)$ is that function of x , which is free from arbitrary constants and obtained $R(x)$ when $\frac{1}{f(D)}R(x)$ is operated by $f(D)$. i.e. $f(D)\frac{1}{f(D)}R(x) = R(x)$.

Therefore, $f(D)$ and $\frac{1}{f(D)}$ are inverse operators, i.e., they cancel each other's effect on the function on which they operate.

Thus the symbol $\frac{1}{D}$ stands for integration, i.e., $\frac{1}{D}$ is denoted by $\int dx$, for instant, $\frac{1}{D} f(x) = \int f(x) dx$.

Similarly, $\frac{1}{D^2} f(x) = \iint f(x) dx dx$, and so on.

Note : $\frac{1}{D} = \int dx = x$; $\frac{1}{D^2} = \iint dx dx = \int x dx = \frac{x^2}{2!}$; $\frac{1}{D^3} = \iiint dx dx dx = \iint x dx dx = \int \frac{x^2}{2} dx = \frac{x^3}{3!}$; , $\frac{1}{D^n} = \frac{x^n}{n!}$.

6.1.6 Particular Integral of $f(D)y = R(x)$

Theorem 6.1.2: Particular Integral of $f(D)y = R(x)$ is $y_p = \frac{1}{f(D)} R(x)$.

6.1.7 Solution of Non-homogeneous Differential Equation of the form $f(D)y = R(x)$

Based on the characteristics of $R(x)$, we consider the differential equation in the following types:

Type-A: When $R(x)$ is of the form x^n in the differential equation $f(D)y = R(x)$.

Type-B: When $R(x)$ is of the form e^{ax} in the differential equation $f(D)y = R(x)$.

Type-C: When $R(x)$ is of the form $\sin ax$ or $\cos ax$ in the differential equation $f(D)y = R(x)$.

Type-D: When $R(x)$ is of the form $x^n \sin ax$ or $x^n \cos ax$ in the differential equation $f(D)y = R(x)$.

Type-E: When $R(x)$ is of the form $e^{ax} f(x)$ in the differential equation $f(D)y = R(x)$.

6.1.8 Solution of the Differential Equation $f(D)y = R(x)$ when $R(x)$ is of the form x^n

Theorem 6.1.3: If $R(x)$ is the function of x only, then

$$\frac{1}{D-\alpha} R(x) = e^{\alpha x} \frac{1}{D} \left\{ e^{-\alpha x} R(x) \right\} = e^{\alpha x} \int e^{-\alpha x} R(x) dx$$

$$\text{and } \frac{1}{D+\alpha} R(x) = e^{-\alpha x} \frac{1}{D} \left\{ e^{\alpha x} R(x) \right\} = e^{-\alpha x} \int e^{\alpha x} R(x) dx$$

Note: When $R(x)$ is given any one of the form $\tan ax$, $\cot ax$, $\sec ax$ or $\operatorname{cosec} ax$, then the above formulae are used in order to determine particular integral.

6.1.9 Working Rule for Finding the Particular Integral of $f(D)y = R(x)$

Let $f(D) = (D-\alpha_1)(D-\alpha_2)(D-\alpha_3) \dots (D-\alpha_n)$. Then resolving into partial fraction, we get

$$\frac{1}{f(D)} = \frac{A_1}{D-\alpha_1} + \frac{A_2}{D-\alpha_2} + \frac{A_3}{D-\alpha_3} + \dots + \frac{A_n}{D-\alpha_n}$$

Now particular integral

$$\begin{aligned} \frac{1}{f(D)} R(x) &= \left[\frac{A_1}{D-\alpha_1} + \frac{A_2}{D-\alpha_2} + \frac{A_3}{D-\alpha_3} + \dots + \frac{A_n}{D-\alpha_n} \right] R(x) \\ &= A_1 \frac{1}{D-\alpha_1} R(x) + A_2 \frac{1}{D-\alpha_2} R(x) + \dots + A_n \frac{1}{D-\alpha_n} R(x) \\ &= A_1 e^{\alpha_1 x} \int e^{-\alpha_1 x} R(x) dx + A_2 e^{\alpha_2 x} \int e^{-\alpha_2 x} R(x) dx + \dots + A_n e^{\alpha_n x} \int e^{-\alpha_n x} R(x) dx \end{aligned}$$

Which can be evaluated and thus the particular integral be found.

Rule-1: In $y_p = \frac{1}{f(D)} R(x)$ if $R(x)$ is of the form x^n , where $n \in \mathbb{N}$ i.e., $y_p = \frac{1}{f(D)} x^n$, then expanding $[f(D)]^{-1}$ as binomial theorem in ascending power of D , obtained the particular integral by operate it.

6.1.10 Some Important Binomial Expansions

- i. $(1-D)^{-1} = 1 + D + D^2 + D^3 + \dots + D^n + \dots$
- ii. $(1+D)^{-1} = 1 - D + D^2 - D^3 + \dots + (-1)^n D^n + \dots$
- iii. $(1-D)^{-2} = 1 + 2D + 3D^2 + 4D^3 + \dots + (n-1)D^n + \dots$
- iv. $(1+D)^{-2} = 1 - 2D + 3D^2 - 4D^3 + \dots + (-1)^n (n-1)D^n + \dots$
- v. $(1-D)^{-3} = 1 + 3D + 6D^2 + 10D^3 + \dots + \frac{n(n+1)}{2} D^{n-1} + \dots$
- vi. $(1+D)^{-3} = 1 - 3D + 6D^2 - 10D^3 + \dots + (-1)^{n-1} \frac{n(n+1)}{2} D^{n-1} + \dots$

Solved Problems:

Example-1 Find the particular integral of $(D+\alpha)y = x^n$, where $n \in \mathbb{N}$.

Solution The particular integral of $(D+\alpha)y = x^n$ is

$$y_p = \frac{1}{D+\alpha} x^n = \frac{1}{\alpha \left(1 + \frac{D}{\alpha}\right)} x^n = \frac{1}{\alpha} \left(1 + \frac{D}{\alpha}\right)^{-1} x^n$$

$$= \frac{1}{\alpha} \left[1 - \frac{D}{\alpha} + \frac{D^2}{\alpha^2} - \dots + (-1)^n \frac{D^n}{\alpha^n} + (-1)^{n+1} \frac{D^{n+1}}{\alpha^{n+1}} + \dots \right] x^n = \frac{1}{\alpha} \left[x^n - \frac{nx^{n-1}}{\alpha} + \frac{n(n-1)x^{n-2}}{\alpha^2} - \dots + (-1)^n \frac{n!}{\alpha^n} \right]$$

Since, $D^{n+1}(x^n) = D(D^n x^n) = D(n!) = 0$, $D^{n+2}(x^n) = 0, \dots$ and so on.

Example-2 Solve the differential equation $\frac{d^2 y}{dx^2} - 3\frac{dy}{dx} + 2y = 4x^2$

Solution Given differential equation can be written as follows

$$(D^2 - 3D + 2)y = 4x^2 \quad \dots (1)$$

The corresponding homogeneous differential equation is

$$(D^2 - 3D + 2)y = 0 \quad \dots (2)$$

Let $y = e^{mx}$ be the trial solution of the differential equation (2), then we get

$$Dy = me^{mx} \text{ and } D^2 y = m^2 e^{mx}$$

Putting these values in equation (1), we obtain $(m^2 - 3m + 2)e^{mx} = 0$.

Since $e^{mx} \neq 0$, the auxiliary equation of the homogeneous differential equation of (2), is

$$m^2 - 3m + 2 = 0$$

$$\therefore m = 1, 2$$

Therefore, the complementary solution of (1) is

$$y_c = c_1 e^x + c_2 e^{2x}$$

Now, the particular integral of the differential equation (1) is

$$y_p = \frac{1}{D^2 - 3D + 2} 4x^2 = \frac{1}{2 \left(1 - \frac{3D}{2} + \frac{D^2}{2}\right)} 4x^2$$

$$= 2 \left\{ 1 - \left(\frac{3D}{2} - \frac{D^2}{2} \right) \right\}^{-1} (x^2)$$

$$= 2 \left[1 + \left(\frac{3D}{2} - \frac{D^2}{2} \right) + \left(\frac{3D}{2} - \frac{D^2}{2} \right)^2 + \dots \right] (x^2)$$

$$= 2 \left[1 + \frac{3D}{2} - \frac{D^2}{2} + \frac{9D^2}{4} - \frac{3D^3}{2} + \frac{D^4}{4} + \dots \right] (x^2)$$

$$= 2 \left[x^2 + \frac{3}{2}(2x) - \frac{1}{2}(2) + \frac{9}{4}(2) - 0 \right]$$

$$= 2 \left[x^2 + 3x + \frac{7}{2} \right] = 2x^2 + 6x + 7$$

Thus, the general solution of the differential equation is

$$y = y_c + y_p = c_1 e^x + c_2 e^{2x} + 2x^2 + 6x + 7 \quad \text{(Ans.)}$$

Example-3 Solve the Differential Equation $(2D^2 + 5D + 2)y = x^2$

Solution Given differential equation is

$$(2D^2 + 5D + 2)y = x^2 \quad \dots (1)$$

The corresponding homogeneous differential equation is $(2D^2 + 5D + 2)y = 0 \quad \dots (2)$

Let $y = e^{mx}$ be a trial solution of the equation (2), then the auxiliary equation of (2) is

$$2m^2 + 5m + 2 = 0 \therefore m = -2, -\frac{1}{2}$$

Therefore, the complementary solution of (1) is

$$y_c = c_1 e^{-x/2} + c_2 e^{-2x}$$

Now, the particular integral of the differential equation (1) is

$$y_p = \frac{1}{2D^2 + 5D + 2} x^2 = \frac{1}{2\left(1 + \frac{5D}{2} + D^2\right)} x^2$$

$$= \frac{1}{2} \left\{ 1 + \left(\frac{5D}{2} + D^2 \right) \right\}^{-1} (x^2)$$

$$= \frac{1}{2} \left[1 - \left(\frac{5D}{2} + D^2 \right) + \left(\frac{5D}{2} + D^2 \right)^2 - \dots \right] (x^2)$$

$$= \frac{1}{2} \left[1 - \frac{5D}{2} - D^2 + \frac{25D^2}{4} + 5D^3 + D^4 - \dots \right] (x^2)$$

$$= \frac{1}{2} \left[x^2 - \frac{5}{2}(2x) - 2 + \frac{25}{4}(2) - 0 \right]$$

$$= \frac{1}{2} \left[x^2 - 5x + \frac{21}{2} \right]$$

Thus, the general solution of the differential equation is $y = y_c + y_p$

$$= c_1 e^{-x/2} + c_2 e^{-2x} + \frac{1}{2} \left(x^2 - 5x + \frac{21}{2} \right) \quad \text{(Ans.)}$$

Example-4 Solve the differential equation $(D^2 - 6D + 9)y = 1 + x + x^2$

Solution Given differential equation is

$$(D^2 - 6D + 9)y = 1 + x + x^2 \quad \dots (1)$$

The corresponding homogeneous differential equation is $(D^2 - 6D + 9)y = 0 \quad \dots (2)$

Let $y = e^{mx}$ be a trial solution of the equation (2), then the auxiliary equation of (2) is

$$m^2 - 6m + 9 = 0 \Rightarrow (m - 3)^2 = 0 \therefore m = 3, 3$$

Therefore, the complementary solution of (1) is

$$y_c = (c_1 + c_2 x) e^{3x}$$

Now, the particular integral of the differential equation (1) is $y_p = \frac{1}{D^2 - 6D + 9} (1 + x + x^2)$

$$= \frac{1}{(D - 3)^2} (1 + x + x^2)$$

$$= \frac{1}{9} \left(1 - \frac{D}{3} \right)^{-2} (1 + x + x^2)$$

$$= \frac{1}{9} \left(1 + 2 \cdot \frac{D}{3} + 3 \cdot \frac{D^2}{3^2} + \dots \right)^{-2} (1 + x + x^2)$$

$$= \frac{1}{9} \left\{ (1 + x + x^2) + \frac{2}{3}(1 + 2x) + \frac{2}{3} + 0 \right\}$$

$$= \frac{1}{27} (3x^2 + 7x + 7)$$

Thus, the G.S. of the differential equation is

$$y = y_c + y_p = (c_1 + c_2 x) e^{3x} + \frac{1}{27} (3x^2 + 7x + 7) \quad \text{(Ans.)}$$

Example-5 Solve the differential equation $(D^2 - 4D + 4)y = x^3$

Solution Given differential equation is

$$(D^2 - 4D + 4)y = x^3 \quad \dots (1)$$

The corresponding homogeneous differential equation is $(D^2 - 4D + 4)y = 0 \quad \dots (2)$

Let $y = e^{mx}$ be a trial solution of the equation (2), then the auxiliary equation of (2) is

$$m^2 - 4m + 4 = 0 \therefore m = 2, 2$$

Therefore, the complementary solution of (1) is

$$y_c = (c_1 + c_2 x) e^{2x}$$

Now, the particular integral of the differential

equation (1) is

$$y_p = \frac{1}{D^2 - 4D + 4} x^3 = \frac{1}{(D - 2)^2} x^3 = \frac{1}{4} \left(1 - \frac{D}{2} \right)^{-2} x^3$$

$$= \frac{1}{4} \left(1 + 2 \cdot \frac{D}{2} + 3 \cdot \frac{D^2}{2^2} + 4 \cdot \frac{D^3}{2^3} + \dots \right) x^3$$

$$= \frac{1}{4} \left(x^3 + 3x^2 + \frac{3}{4} \cdot 6x + \frac{6}{2} + 0 \right) = \frac{1}{8} (2x^3 + 6x^2 + 9x + 6)$$

Thus, the general solution of the differential equation is $y = y_c + y_p$

$$= (c_1 + c_2 x) e^{2x} + \frac{1}{8} (2x^3 + 6x^2 + 9x + 6) \quad \text{(Ans.)}$$

Section 6.2

Solution of the differential equation $f(D)y = R(x)$ when $R(x)$ is of the form e^{ax}

6.2.1 Solution of the Differential Equation $f(D)y = R(x)$ when $R(x)$ is of the form e^{ax}

Theorem 6.2.1: For all $n \in \mathbb{N}$, $D^n e^{ax} = a^n e^{ax}$.

Corollary-1: $f(D)e^{ax} = f(a)e^{ax}$

Corollary-2: For all $n \in \mathbb{N}$, $(D-a)^n (x^n e^{ax}) = n! e^{ax}$.

Theorem 6.2.2: If $f(a) \neq 0$, then $\frac{1}{f(D)} e^{ax} = \frac{1}{f(a)} e^{ax}$.

Theorem 6.2.3: If $f(a) = 0$, but $f'(a) \neq 0$ then $\frac{1}{f(D)} e^{ax} = x \frac{1}{f'(D)} e^{ax} = x \frac{1}{f'(a)} e^{ax}$.

Theorem 6.2.4: If $f(a) = 0$ and $f'(a) = 0$ but $f''(a) \neq 0$ then $\frac{1}{f(D)} e^{ax} = x^2 \frac{1}{f''(D)} e^{ax} = x^2 \frac{1}{f''(a)} e^{ax}$.

Theorem 6.2.5: If $f(a) = 0$, $f'(a) = 0$, $f''(a) = 0, \dots, f^{(n)}(a) = 0$, but $f^{(n+1)}(a) \neq 0$, then

$$\frac{1}{f(D)} e^{ax} = x^{n+1} \frac{1}{f^{(n+1)}(a)} e^{ax}.$$

Corollary-3: If $f(0) \neq 0$, then $\frac{1}{f(D)} = \frac{1}{f(D)} e^{0 \cdot x} = \frac{1}{f(0)} e^{0 \cdot x} = \frac{1}{f(0)}$.

Examples:

$$(i) \quad \frac{1}{D^2 + 1} e^{2x} = \frac{1}{2^2 + 1} e^{2x} = \frac{1}{5} e^{2x}$$

$$(ii) \quad \frac{1}{D^2 - 1} e^x = x \frac{1}{2D} e^x = x \cdot \frac{1}{2 \cdot 1} e^x = \frac{1}{2} x e^x$$

$$(iii) \quad \frac{1}{(D+1)^2} e^{-x} = x \frac{1}{2(D+1)} e^{-x} = x^2 \cdot \frac{1}{2} e^{-x} = \frac{1}{2} x^2 e^{-x}$$

$$(iv) \quad \frac{1}{D^2 - 3D + 2} = \frac{1}{D^2 - 3D + 2} e^{0 \cdot x} = \frac{1}{0^2 - 3 \cdot 0 + 2} = \frac{1}{2}$$

Solved Problems:

Example-1 Solve the differential equation $(D^2 - 3D + 2)y = e^{5x}$

Solution Given differential equation is

$$(D^2 - 3D + 2)y = e^{5x} \quad \dots (1)$$

Let $y = e^{mx}$ be a trial solution of the equation (1), then the auxiliary equation of the corresponding homogeneous equation of (1) is

$$m^2 - 3m + 2 = 0$$

$$\therefore m = 1, 2$$

Therefore, the complementary function

$$y_c = c_1 e^x + c_2 e^{2x}.$$

Now, the particular integral of the differential equation (1) is

$$\begin{aligned} y_p &= \frac{1}{D^2 - 3D + 2} e^{5x} \\ &= \frac{1}{5^2 - 3 \cdot 5 + 2} e^{5x} = \frac{1}{12} e^{5x} \end{aligned}$$

$\therefore$ The general solution is $y = y_c + y_p$

$$\therefore y = c_1 e^x + c_2 e^{2x} + \frac{1}{12} e^{5x} \quad (\text{Ans.})$$

Example-2 Solve the differential equation $(D^2 + 6D + 5)y = 2e^x + 10e^{5x}$

Solution Given differential equation is

$$(D^2 + 6D + 5)y = 2e^x + 10e^{5x} \quad \dots (1)$$

Let $y = e^{mx}$ be a trial solution of the equation (1), then the auxiliary equation of the corresponding homogeneous equation of (1) is

$$m^2 + 6m + 5 = 0$$

$$\therefore m = -1, -5$$

Therefore, the complementary function

$$y_c = c_1 e^{-x} + c_2 e^{-5x}.$$

Now, the particular integral of the differential

equation (1) is

$$\begin{aligned} y_p &= \frac{1}{D^2 + 6D + 5} (2e^x + 10e^{5x}) \\ &= 2 \frac{1}{D^2 + 6D + 5} e^x + 10 \frac{1}{D^2 + 6D + 5} e^{5x} \\ &= 2 \frac{1}{1^2 + 6 \cdot 1 + 5} e^x + 10 \frac{1}{5^2 + 6 \cdot 5 + 5} e^{5x} \\ &= \frac{1}{6} e^x + \frac{1}{6} e^{5x} = \frac{1}{6} (e^x + e^{5x}) \end{aligned}$$

$\therefore$ The general solution is $y = y_c + y_p$

$$= c_1 e^{-x} + c_2 e^{-5x} + \frac{1}{6} (e^x + e^{5x}) \quad \text{(Ans.)}$$

Example-3 Solve the differential equation $(D^2 + 4D + 4)y = 3e^{-x}$, $y(0) = 3$ and $y'(0) = 1$

Solution Given differential equation is

$$(D^2 + 4D + 4)y = 3e^{-x} \quad \dots (1)$$

Let $y = e^{mx}$ be a trial solution of the equation (1), then the auxiliary equation of the corresponding homogeneous equation of (1) is

$$m^2 + 4m + 4 = 0$$

$$\therefore m = -2, -2$$

Therefore, the complementary function

$$y_c = (c_1 + c_2 x) e^{-2x}$$

Now, the particular integral of the differential equation (1) is

$$\begin{aligned} y_p &= \frac{1}{D^2 + 4D + 4} 3e^{-x} = 3 \frac{1}{(D + 2)^2} e^{-x} \\ &= 3 \frac{1}{(-1 + 2)^2} e^{-x} = 3e^{-x} \end{aligned}$$

$\therefore$ General the solution is $y = y_c + y_p$

$$\therefore y(x) = (c_1 + c_2 x) e^{-2x} + 3e^{-x}$$

Also, we have $y'(x) = c_2 e^{-2x} - 2(c_1 + c_2 x) e^{-2x} - 3e^{-x}$

Using initial condition $y(0) = 3$ in equation (2), we get

$$y(0) = c_1 + 3 = 3$$

$$\therefore c_1 = 0$$

Also using $y'(0) = 1$ in equation (3), we get

$$y'(0) = 1 = c_2 - 2c_1 - 3$$

$$\Rightarrow c_2 = 1 + 3 + 2c_1 = 4 + 2 \cdot 0$$

$$\therefore c_2 = 4$$

Putting these values in equation (2), we obtained the desired solution

$$y(x) = 4xe^{-2x} + 3e^{-x} \quad \text{(Ans.)}$$

Example-4 Solve the differential equation $(D^2 - 3D + 2)y = e^x$, $y(0) = 3$ and $y'(0) = 3$

Solution Given differential equation is

$$(D^2 - 3D + 2)y = e^x \quad \dots (1)$$

Let $y = e^{mx}$ be a trial solution of the equation (1), then the auxiliary equation of the corresponding homogeneous equation of (1) is $m^2 - 3m + 2 = 0 \Rightarrow (m - 1)(m - 2) = 0 \therefore m = 1, 2$

Therefore, the complementary function $y_c = c_1 e^x + c_2 e^{2x}$

Now, the particular integral of the differential equation (1) is

$$y_p = \frac{1}{D^2 - 3D + 2} e^x = x \frac{1}{2D - 3} e^x = x \frac{1}{2 \cdot 1 - 3} e^x = -xe^x$$

$\therefore$ General the solution is $y = y_c + y_p$

$$\therefore y(x) = c_1 e^x + c_2 e^{2x} - xe^x \quad \dots (2)$$

Also, we have $y'(x) = c_1 e^x + 2c_2 e^{2x} - x e^x - e^x$... (3)

Using initial condition $y(0) = 3$ in equation (2), we get

$$y(0) = c_1 + c_2 = 3$$

$$\Rightarrow c_1 = 3 - c_2$$

Also using $y'(0) = 3$ in equation (3), we get

$$y'(0) = c_1 + 2c_2 - 1 = 3$$

$$\Rightarrow 3 - c_2 + 2c_2 - 1 = 3$$

$$\therefore c_2 = 1 \text{ and } c_1 = 2$$

Putting these values in equation (2), we obtained the desired solution

$$y(x) = (2 - x)e^x + e^{2x} \text{ (Ans.)}$$

Section 6.3

Solution of the differential equation $f(D)y = R(x)$ when $R(x)$ is of the form $\sin ax$ or $\cos ax$

6.3.1 Solution of the Differential Equation $f(D)y = R(x)$ when $R(x)$ is of the form $\sin ax$ or $\cos ax$

Theorem 6.3.1: For all $n \in \mathbb{N}$, (i) $(D^2)^n \sin ax = (-a^2)^n \sin ax$ and (ii) $(D^2)^n \cos ax = (-a^2)^n \cos ax$

Corollary-4: $f(D^2)\sin ax = f(-a^2)\sin ax$ and $f(D^2)\cos ax = f(-a^2)\cos ax$

Theorem 6.3.2: If $f(-a^2) \neq 0$, then (i) $\frac{1}{f(D^2)}\sin ax = \frac{1}{f(-a^2)}\sin ax$ and

$$(ii) \frac{1}{f(D^2)}\cos ax = \frac{1}{f(-a^2)}\cos ax$$

Theorem 6.3.3: If $f(-a^2) = 0$, but $f'(-a^2) \neq 0$ then

$$(i) \frac{1}{f(D^2)}\sin ax = x \frac{1}{f'(D^2)}\sin ax = x \frac{1}{f'(-a^2)}\sin ax \text{ and}$$

$$(ii) \frac{1}{f(D^2)}\cos ax = x \frac{1}{f'(D^2)}\cos ax = x \frac{1}{f'(-a^2)}\cos ax$$

Theorem 6.3.4: If $f(-a^2) = 0$ and $f'(-a^2) = 0$ but $f''(-a^2) \neq 0$ then

$$(i) \frac{1}{f(D^2)}\sin ax = x^2 \frac{1}{f''(D^2)}\sin ax = x^2 \frac{1}{f''(-a^2)}\sin ax \text{ and}$$

$$(ii) \frac{1}{f(D^2)}\cos ax = x^2 \frac{1}{f''(D^2)}\cos ax = x^2 \frac{1}{f''(-a^2)}\cos ax$$

Theorem 6.3.5: If $f(-a^2) = 0$, $f'(-a^2) = 0$, $f''(-a^2) = 0$, $\dots$, $f^{(n)}(-a^2) = 0$, but $f^{(n+1)}(-a^2) \neq 0$, then

$$(i) \frac{1}{f(D^2)}\sin ax = x^{n+1} \frac{1}{f^{(n+1)}(D^2)}\sin ax = x^{n+1} \frac{1}{f^{(n+1)}(-a^2)}\sin ax \text{ and}$$

$$(ii) \frac{1}{f(D^2)}\cos ax = x^{n+1} \frac{1}{f^{(n+1)}(D^2)}\cos ax = x^{n+1} \frac{1}{f^{(n+1)}(-a^2)}\cos ax$$

Examples:

$$(i) \frac{1}{D^2 + 9}\sin 2x = \frac{1}{-2^2 + 9}\sin 2x = \frac{1}{5}\sin 2x;$$

$$(ii) \frac{1}{D^2 + 1}\cos x = x \frac{1}{2D}\cos x = \frac{x}{2} \int \cos x dx = \frac{x}{2}\sin x$$

Solved Problems:

Example-1 Solve the differential equations:

$$(i) (D^2 + a^2)y = \cos ax$$

$$(ii) (D^2 + a^2)y = \sin ax$$

Solution

(i) Given differential equation is

$$(D^2 + a^2)y = \cos ax \quad \dots (1)$$

Let $y = e^{mx}$ be a trial solution of the equation (1), then the auxiliary equation of the corresponding homogeneous equation of (1) is

$$m^2 + a^2 = 0 \Rightarrow m^2 = -a^2 \Rightarrow m^2 = i^2 a^2 \quad \therefore m = \pm ai$$

Therefore, the complementary function $y_c = A \cos ax + B \sin ax$.

Now, the particular integral of the differential equation (1) is

$$\begin{aligned} y_p &= \frac{1}{D^2 + a^2} \cos ax \\ &= x \frac{1}{2D} \cos ax = \frac{x}{2} \int \cos ax \, dx = \frac{x}{2a} \sin ax \end{aligned}$$

$\therefore$ The general solution is $y = y_c + y_p$

$$\therefore y = A \cos ax + B \sin ax + \frac{x}{2a} \sin ax \quad (\text{Ans.})$$

(ii) **Try yourself** [Ans. $y = A \cos ax + B \sin ax - \frac{x}{2a} \cos ax$]

Example-2 Solve the differential equations:

(i) $(D^2 + 1)y = \cos^2 x$

(ii) $(D^2 + 1)y = \sin^2 x$

Solution

(i) Given differential equation is

$$(D^2 + 1)y = \cos^2 x \quad \dots (1)$$

Let $y = e^{mx}$ be a trial solution of the equation (1), then the auxiliary equation of the corresponding homogeneous equation of (1) is

$$m^2 + 1 = 0 \Rightarrow m^2 = -1 \quad \therefore m = \pm i$$

Therefore, the complementary function $y_c = A \cos x + B \sin x$.

Now, the particular integral of the differential equation (1) is

$$\begin{aligned} y_p &= \frac{1}{D^2 + 1} \cos^2 x = \frac{1}{2} \frac{1}{D^2 + 1} (1 + \cos 2x) = \frac{1}{2} \left\{ \frac{1}{D^2 + 1} (1) + \frac{1}{D^2 + 1} (\cos 2x) \right\} \\ &= \frac{1}{2} \left\{ \frac{1}{0^2 + 1} e^{0 \cdot x} + \frac{1}{-2^2 + 1} (\cos 2x) \right\} = \frac{1}{2} \left(1 - \frac{1}{3} \cos 2x \right) = \frac{1}{6} (3 - \cos 2x) \end{aligned}$$

$\therefore$ The general solution is $y = y_c + y_p$

$$\therefore y = A \cos x + B \sin x + \frac{1}{6} (3 - \cos 2x) \quad (\text{Ans.})$$

(ii) **[Try yourself]** [Ans. $y = A \cos x + B \sin x + \frac{1}{6} (3 + \cos 2x)$]

Example-3 Solve the differential equation $(D^2 + 2D + 2)y = 10 \sin 4x$

Solution Given differential equation is

$$(D^2 + 2D + 2)y = 10 \sin 4x \quad \dots (1)$$

Let $y = e^{mx}$ be a trial solution of the equation (1), then the auxiliary equation of the corresponding homogeneous equation of (1) is

$$\begin{aligned} m^2 + 2m + 2 &= 0 \Rightarrow m^2 + 2m + 1 = -1 \Rightarrow (m+1)^2 = i^2 \quad [\because i^2 = -1] \\ &\Rightarrow m+1 = \pm i \quad \therefore m = -1 \pm i \end{aligned}$$

Therefore, the complementary function $y_c = e^{-x}(A \cos x + B \sin x)$.

Now, the particular integral of the differential equation (1) is

$$\begin{aligned} y_p &= \frac{1}{D^2 + 2D + 2} 10 \sin 4x \\ &= 10 \frac{1}{-4^2 + 2D + 2} \sin 4x \\ &= 10 \frac{1}{2D - 14} \sin 4x \\ &= 5 \frac{(D + 7)}{(D - 7)(D + 7)} \sin 4x \\ &= 5 \frac{(D + 7)}{(D^2 - 49)} \sin 4x \end{aligned}$$

$$\begin{aligned} &= 5 \frac{(D + 7)}{(-4^2 - 49)} \sin 4x \\ &= 5 \frac{(4 \cos 4x + 7 \sin 4x)}{-65} \\ &= -\frac{1}{13} (4 \cos 4x + 7 \sin 4x) \end{aligned}$$

$\therefore$ The general solution is $y = y_c + y_p$

$$\therefore y = e^{-x}(A \cos x + B \sin x) - \frac{1}{13}(4 \cos 4x + 7 \sin 4x) \quad (\text{Ans.})$$

Example-4 Solve the differential equations:

(i) $(D^2 - 4)y = e^x - \sin x$ (ii) $(D^3 - 3D^2 + 4D - 2)y = e^x + \cos x$

Solution

(i) Given differential equation is

$$(D^2 - 4)y = e^x - \sin x \quad \dots (1)$$

Let $y = e^{mx}$ be a trial solution of the equation (1), then the auxiliary equation of the corresponding homogeneous equation of (1) is

$$m^2 - 4 = 0$$

$$\therefore m = \pm 2$$

Therefore, the complementary function $y_c = c_1 e^{2x} + c_2 e^{-2x}$.

Now, the particular integral of the differential equation (1) is

$$\begin{aligned} y_p &= \frac{1}{D^2 - 4} (e^x - \sin x) \\ &= \frac{1}{D^2 - 4} e^x - \frac{1}{D^2 - 4} \sin x \\ &= \frac{1}{1^2 - 4} e^x - \frac{1}{-1^2 - 4} \sin x \\ &= \frac{1}{5} \sin x - \frac{1}{3} e^x \end{aligned}$$

$\therefore$ The general solution is $y = y_c + y_p$

$$\therefore y = c_1 e^{2x} + c_2 e^{-2x} + \frac{1}{5} \sin x - \frac{1}{3} e^x \quad (\text{Ans.})$$

(ii) Given differential equation is

$$(D^3 - 3D^2 + 4D - 2)y = e^x + \cos x \quad \dots (1)$$

Let $y = e^{mx}$ be a trial solution of the equation (1), then the auxiliary equation of the corresponding homogeneous equation of (1) is $m^3 - 3m^2 + 4m - 2 = 0$

$$\Rightarrow m^2(m - 1) - 2m(m - 1) + 2(m - 1) = 0$$

$$\Rightarrow (m - 1)(m^2 - 2m + 2) = 0$$

$$\Rightarrow (m - 1)\{(m - 1)^2 + 1\} = 0$$

$$\Rightarrow m = 1 \text{ and } (m - 1)^2 = -1 = i^2$$

$$\therefore m = 1 \text{ and } m = 1 \pm i$$

Therefore, the complementary function $y_c = e^x(c_1 \cos x + c_2 \sin x)$.

Now, the particular integral of the differential equation

$$(1) \text{ is } y_p = \frac{1}{D^3 - 3D^2 + 4D - 2} (e^x + \cos x)$$

$$= \frac{1}{D^3 - 3D^2 + 4D - 2} e^x + \frac{1}{D^2 \cdot D - 3D^2 + 4D - 2} \cos x$$

$$= x \frac{1}{3D^2 - 6D + 4} e^x + \frac{1}{(-1^2) \cdot D - 3(-1^2) + 4D - 2} \cos x$$

$$= x \frac{1}{3 \cdot 1^2 - 6 \cdot 1 + 4} e^x + \frac{1}{3D + 2} \cos x$$

$$= x e^x + \frac{3D + 2}{9D^2 - 4} \cos x = x e^x + \frac{3D + 2}{9(-1^2) - 4} \cos x$$

$$= x e^x - \frac{1}{13} (-3 \sin x + 2 \cos x)$$

$\therefore$ The general solution is $y = y_c + y_p$

$$\therefore y = e^x(c_1 \cos x + c_2 \sin x) + x e^x + \frac{1}{13}(3 \sin x - 2 \cos x) \quad (\text{Ans.})$$

Example-5 Solve the differential equations:

(i) $(D^2 - 2D + 1)y = x + \sin x$

(ii) $(D^2 + 2D + 3)y = x^2 + \cos x$

Solution

(i) Given differential equation is

$$(D^2 - 2D + 1)y = x + \sin x \quad \dots (1)$$

Let $y = e^{mx}$ be a trial solution of the equation (1), then the auxiliary equation of the corresponding homogeneous equation of (1) is

$$\begin{aligned} m^2 - 2m + 1 &= 0 \\ \Rightarrow (m-1)^2 &= 0 \\ \therefore m &= 1, 1 \end{aligned}$$

Therefore, the complementary function $y_c = (c_1 + c_2x)e^x$.

Now, the particular integral of the differential equation (1) is

$$\begin{aligned} y_p &= \frac{1}{D^2 - 2D + 1}(x + \sin x) \\ &= \frac{1}{(D-1)^2}x + \frac{1}{D^2 - 2D + 1}\sin x \\ &= (1-D)^{-2}x + \frac{1}{-1^2 - 2D + 1}\sin x \\ &= (1 + 2D + 3D^2 + \dots)x - \frac{1}{2D}\sin x \\ &= (x + 2 \cdot 1 + 0) - \frac{1}{2}\int \sin x \, dx \\ &= (x + 2) + \frac{1}{2}\cos x \end{aligned}$$

$\therefore$ The general solution is $y = y_c + y_p$

$$\therefore y = (c_1 + c_2x)e^x + (x + 2) + \frac{1}{2}\cos x \quad \text{(Ans.)}$$

(ii) Given differential equation is

$$(D^2 + 2D + 3)y = x^2 + \cos x \quad \dots (1)$$

Let $y = e^{mx}$ be a trial solution of the equation (1), then the auxiliary equation of the corresponding homogeneous equation of (1) is

$$\begin{aligned} m^2 + 2m + 3 &= 0 \Rightarrow m^2 + 2m + 1 = -2 \Rightarrow (m+1)^2 = -2 \\ \Rightarrow (m+1) &= \pm i\sqrt{2} \quad \therefore m = -1 \pm i\sqrt{2} \end{aligned}$$

Therefore, the complementary function

$$y_c = e^{-x}(c_1 \cos \sqrt{2}x + c_2 \sin \sqrt{2}x).$$

Now, the particular integral of the differential equation (1) is

$$\begin{aligned} y_p &= \frac{1}{D^2 + 2D + 3}(x^2 + \cos x) \\ &= \frac{1}{3\left(1 + \frac{2D}{3} + \frac{D^2}{3}\right)}x^2 + \frac{1}{D^2 + 2D + 3}\cos x \\ &= \frac{1}{3}\left\{1 + \left(\frac{2D}{3} + \frac{D^2}{3}\right)\right\}^{-1}x^2 + \frac{1}{-1^2 + 2D + 3}\cos x \\ &= \frac{1}{3}\left\{1 - \left(\frac{2D}{3} + \frac{D^2}{3}\right) + \left(\frac{2D}{3} + \frac{D^2}{3}\right)^2 - \dots\right\}^{-1}x^2 + \frac{1}{2(D+1)}\cos x \\ &= \frac{1}{3}\left(1 - \frac{2D}{3} - \frac{D^2}{3} + \frac{4D^2}{9} - \dots\right)x^2 + \frac{(D-1)}{2(D^2-1)}\cos x \\ &= \frac{1}{3}\left(x^2 - \frac{2}{3} \cdot 2x - \frac{2}{3} + \frac{4 \cdot 2}{9} - 0\right) + \frac{(D-1)}{2(-1^2-1)}\cos x \\ &= \frac{1}{3}\left(x^2 - \frac{4}{3}x - \frac{2}{3} + \frac{8}{9}\right) - \frac{1}{4}(\sin x - \cos x) \\ &= \frac{1}{27}(9x^2 - 12x + 2) - \frac{1}{4}(\sin x - \cos x) \end{aligned}$$

$\therefore$ The general solution is $y = y_c + y_p$

$$\therefore y = e^{-x}(c_1 \cos \sqrt{2}x + c_2 \sin \sqrt{2}x) + \frac{1}{27}(9x^2 - 12x + 2) - \frac{1}{4}(\sin x - \cos x) \quad \text{(Ans.)}$$

Section 6.4

Solution of the differential equation $f(D)y = R(x)$ when $R(x)$ is of the form $e^{ax}V$

6.4.1 Solution of the Differential Equation $f(D)y = R(x)$ when $R(x)$ is of the form $e^{ax}V$

In the differential equation $f(D)y = e^{ax}V$, when $V(x)$ is of the form $\sin ax$ or $\cos ax$ or x^n , then we can find the particular integral using the following theorems.

Theorem-6.4.1: If V is a function of x , then or $\frac{1}{f(D)}e^{ax}V = e^{ax} \frac{1}{f(D+a)}V$.

Theorem-6.4.2: If V is a function of x , then or $\frac{1}{f(D)}xV = x \frac{1}{f(D)}V - \frac{f'(D)}{\{f(D)\}^2}V$.

Example-1 Solve the differential equations:

(i) $(D^2 - 2D + 5)y = e^{2x} \sin x$

(ii) $(D^2 - 2D + 5)y = e^{2x} \cos x$

Solution (i) Given differential equation is $(D^2 - 2D + 5)y = e^{2x} \sin x$ (1)

Let $y = e^{mx}$ be a trial solution of the homogeneous equation of (1), then the auxiliary equation of the corresponding homogeneous equation of (1) is

$$\begin{aligned} m^2 - 2m + 5 &= 0 \\ \Rightarrow m &= \frac{2 \pm \sqrt{4 - 20}}{2} = \frac{2 \pm \sqrt{-16}}{2} = \frac{2 \pm 2\sqrt{4i^2}}{2} \\ \therefore m &= 1 \pm 2i \end{aligned}$$

Therefore, the complementary function $y_c = e^x (A \cos 2x + B \sin 2x)$.

Now, the particular integral of the differential equation (1) is

$\begin{aligned} y_p &= \frac{1}{D^2 - 2D + 5} e^{2x} \sin x \\ &= e^{2x} \frac{1}{(D+2)^2 - 2(D+2) + 5} \sin x \\ &= e^{2x} \frac{1}{D^2 + 4D + 4 - 2D - 4 + 5} \sin x \\ &= e^{2x} \frac{1}{D^2 + 2D + 5} \sin x \\ &= e^{2x} \frac{1}{-1^2 + 2D + 5} \sin x \\ &= e^{2x} \frac{1}{2(D+2)} \sin x \\ &= e^{2x} \frac{(D-2)}{2(D+2)(D-2)} \sin x \end{aligned}$	$\begin{aligned} &= e^{2x} \frac{(D-2)}{2(D^2 - 4)} \sin x \\ &= e^{2x} \frac{(D-2)}{2(-1^2 - 4)} \sin x \\ &= -\frac{e^{2x}}{10} (D-2) \sin x \\ &= -\frac{e^{2x}}{10} (\cos x - 2 \sin x) \\ &= \frac{e^{2x}}{10} (2 \sin x - \cos x) \\ \therefore \text{The general solution is } y &= y_c + y_p \\ \therefore y &= e^x (A \cos 2x + B \sin 2x) + \frac{e^{2x}}{10} (2 \sin x - \cos x) \text{ (Ans.)} \end{aligned}$
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Example-2 Solve the differential equations:

(i) $(D^2 - 2D + 1)y = e^x \sin x$ (Ans.) $y = (c_1 + c_2 x)e^x - e^x \sin x$

(ii) $(D^2 - 2D + 1)y = e^x \cos x$ (Ans.) $y = (c_1 + c_2 x)e^x - e^x \cos x$

Example-3 Solve the differential equations:

(i) $(D^4 - 1)y = e^x \cos x$ (Ans.) $y = c_1 e^x + c_2 e^{-x} + c_3 \cos x + c_4 \sin x - \frac{1}{5} e^x \cos x$

(ii) $(D^4 - 1)y = e^x \sin x$ (Ans.) $y = c_1 e^x + c_2 e^{-x} + c_3 \cos x + c_4 \sin x - \frac{1}{5} e^x \sin x$

Example-4 Solve the differential equations:

(i) $(D^2 - 4D + 4)y = x^2 e^{2x}$

(ii) $(D^2 - 9)y = 2x e^{3x}$ [Try yourself] (Ans.) $y = c_1 e^{3x} + c_2 e^{-3x} + \frac{1}{108} e^{3x} (18x^2 - 6x + 1)$

(iii) $(D^2 + 1)y = e^x (x^2 - 1)$

Solution (i) Given differential equation is $(D^2 - 4D + 4)y = x^2 e^{2x}$ (1)

Let $y = e^{mx}$ be a trial solution of the homogeneous equation of (1), then the auxiliary equation of the corresponding homogeneous equation of (1) is

$$m^2 - 4m + 4 = 0 \quad \therefore m = 2, 2$$

Therefore, the complementary function $y_c = (c_1 + c_2 x)e^{2x}$.

Now, the particular integral of the differential equation (1) is

$$\begin{aligned} y_p &= \frac{1}{D^2 - 4D + 4} x^2 e^{2x} = \frac{1}{(D-2)^2} x^2 e^{2x} = e^{2x} \frac{1}{(D+2-2)^2} x^2 \\ &= e^{2x} \frac{1}{D^2} x^2 = e^{2x} \int \int x^2 dx dx = \frac{e^{2x}}{3} \int x^3 dx = \frac{1}{12} x^4 e^{2x} \end{aligned}$$

$\therefore$ The general solution is $y = y_c + y_p$

$$\therefore y = (c_1 + c_2 x)e^{2x} + \frac{1}{12} x^4 e^{2x} \text{ (Ans.)}$$

(iii) Given differential equation is

$$(D^2 + 1)y = e^x (x^2 - 1) \quad \text{.....(1)}$$

Let $y = e^{mx}$ be a trial solution of the homogeneous equation of (1), then the auxiliary equation of the corresponding homogeneous equation of (1) is

$$m^2 + 1 = 0 \quad \therefore m = \pm i$$

Therefore, the complementary function $y_c = c_1 \cos x + c_2 \sin x$.

Now, the particular integral of the differential equation (1) is

$$\begin{aligned} y_p &= \frac{1}{D^2 + 1} e^x (x^2 - 1) \\ &= e^x \frac{1}{(D+1)^2 + 1} (x^2 - 1) \\ &= e^x \frac{1}{D^2 + 2D + 2} (x^2 - 1) \\ &= \frac{1}{2} e^x \frac{1}{1 + D + \frac{D^2}{2}} (x^2 - 1) \\ &= \frac{1}{2} e^x \left\{ 1 + \left(D + \frac{D^2}{2} \right) \right\}^{-1} (x^2 - 1) \end{aligned} \quad \left| \begin{aligned} &= \frac{1}{2} e^x \left\{ 1 - \left(D + \frac{D^2}{2} \right) + \left(D + \frac{D^2}{2} \right)^2 - \dots \right\} (x^2 - 1) \\ &= \frac{1}{2} e^x \left\{ 1 - D - \frac{D^2}{2} + D^2 + D^3 + \dots \right\} (x^2 - 1) \\ &= \frac{1}{2} e^x \left\{ (x^2 - 1) - 2x - \frac{1}{2} \cdot 2 + 2 + 0 \right\} \\ &= \frac{1}{2} x e^x (x - 2) \end{aligned} \right.$$

$\therefore$ The general solution is $y = y_c + y_p$

$$\therefore y = c_1 \cos x + c_2 \sin x + \frac{1}{2} x e^x (x - 2) \text{ (Ans.)}$$

Example-5 Solve the differential equations:

(i) $(D^3 - D^2 + 4D - 4)y = 68e^x \sin 2x$

(ii) $(D^3 - D^2 + 4D - 4)y = 68e^x \cos 2x$

Solution (i) Given differential equation is $(D^3 - D^2 + 4D - 4)y = 68e^x \sin 2x$ (1)

Let $y = e^{mx}$ be a trial solution of the homogeneous equation of (1), then the auxiliary equation of the corresponding homogeneous equation of (1) is

$$m^3 - m^2 + 4m - 4 = 0 \quad \Rightarrow (m-1)(m^2 + 4) = 0 \quad \therefore m = 1, \pm 2i$$

Therefore, the complementary function $y_c = c_1 e^x + c_2 \cos 2x + c_3 \sin 2x$.

Now, the particular integral of the differential equation (1) is

$y_p = \frac{1}{D^3 - D^2 + 4D - 4} 68e^x \sin 2x$ $= 68 \frac{1}{(D-1)(D^2 + 4)} e^x \sin 2x$ $= 68e^x \frac{1}{(D+1-1)\{(D+1)^2 + 4\}} \sin 2x$ $= 68e^x \frac{1}{D(D^2 + 2D + 5)} \sin 2x$ $= 68e^x \frac{1}{D^2 \cdot D + 2D^2 + 5D} \sin 2x$ $= 68e^x \frac{1}{(-2^2) \cdot D + 2(-2^2) + 5D} \sin 2x$	$= 68e^x \frac{1}{D-8} \sin 2x$ $= 68e^x \frac{(D+8)}{D^2 - 8^2} \sin 2x$ $= 68e^x \frac{(D+8)}{-2^2 - 8^2} \sin 2x$ $= -e^x (2 \cos 2x + 8 \sin 2x)$ $\therefore \text{The general solution is } y = y_c + y_p$ $\therefore y = c_1 e^x + c_2 \cos 2x + c_3 \sin 2x - e^x (2 \cos 2x + 8 \sin 2x)$ (Ans.)
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(ii) **[Try yourself]**

Section 6.5

Solution of the differential equation $f(D)y = R(x)$ when $R(x)$ is of the form $x^n \sin ax$ or $x^n \cos ax$

6.5.1 Solution of the Differential Equation $f(D)y = R(x)$ when $R(x)$ is of the form $x^n \sin ax$ or $x^n \cos ax$

Working rule to determine the value of the particular integral of the form

$$y_p = \frac{1}{f(D)} x^n \cos ax \text{ or } y_p = \frac{1}{f(D)} x^n \sin ax :$$

Step-1: Put $X = \frac{1}{f(D)} x^n \cos ax$ and $Y = \frac{1}{f(D)} x^n \sin ax$

Step-2: Then set $X + iY = \frac{1}{f(D)} x^n \cos ax + i \frac{1}{f(D)} x^n \sin ax$

$$= \frac{1}{f(D)} x^n (\cos ax + i \sin ax) = \frac{1}{f(D)} x^n e^{iax}$$

Step-3: Then according theorem- 6.4.1 write $\frac{1}{f(D)} x^n e^{iax} = e^{iax} \frac{1}{f(D+ia)} x^n$ and then find the value of this, i.e., value of $X + iY$.

Step-4: Finally equate to the real and imaginary parts from both sides of the previous value obtained in step-3 and hence find the desired solution.

Example-1 Solve the differential equations:

(i) $(D^2 + 1)y = x \sin x$

(ii) $(D^2 + 1)y = x \cos x$

Solution (i) Given differential equation is $(D^2 + 1)y = x \sin x$ (1)

Let $y = e^{mx}$ be a trial solution of the equation (1), then the auxiliary equation of the corresponding homogeneous equation of (1) is

$$m^2 + 1 = 0 \quad \therefore m = \pm i$$

Therefore, the complementary function $y_c = A \cos x + B \sin x$.

Now, the particular integral of the differential equation (1) is $y_p = \frac{1}{D^2 + 1} x \sin x$

Now, we put $X = \frac{1}{D^2 + 1} x \cos x$ and $Y = \frac{1}{D^2 + 1} x \sin x$, then we have

$$\begin{aligned} X + iY &= \frac{1}{D^2 + 1} x (\cos x + i \sin x) \\ &= \frac{1}{D^2 + 1} x e^{ix} = e^{ix} \frac{1}{(D+i)^2 + 1} x \\ &= e^{ix} \frac{1}{D^2 + 2Di + i^2 + 1} x \\ &= e^{ix} \frac{1}{2Di \left(1 + \frac{D}{2i}\right)} x \end{aligned} \quad \left| \begin{aligned} &= e^{ix} \frac{1}{D^2 + 2Di} x \\ &= e^{ix} \frac{1}{2Di} \left(1 + \frac{D}{2i}\right)^{-1} x \\ &= e^{ix} \frac{1}{2Di} \left(1 - \frac{D}{2i} + \frac{D^2}{4i^2} - \dots\right) x \\ &= e^{ix} \frac{1}{2Di} \left(x - \frac{1}{2i} + 0\right) \end{aligned} \right. \quad \left| \begin{aligned} &= e^{ix} \frac{1}{2i} \left(\frac{x^2}{2} - \frac{x}{2i}\right) \\ &= \frac{1}{2i} (\cos x + i \sin x) \left(\frac{x^2}{2} - \frac{x}{2i}\right) \\ &= (\cos x + i \sin x) \left(\frac{x^2}{4i} - \frac{x}{4i^2}\right) \\ &= (\cos x + i \sin x) \left(\frac{x}{4} - i \frac{x^2}{4}\right) \end{aligned} \right.$$

$$\therefore X + iY = \left(\frac{x}{4} \cos x + \frac{x^2}{4} \sin x \right) + i \left(\frac{x}{4} \sin x - \frac{x^2}{4} \cos x \right)$$

Equating real and imaginary parts from both sides, we get

$$X = \frac{x}{4} \cos x + \frac{x^2}{4} \sin x \quad \text{and} \quad Y = \frac{x}{4} \sin x - \frac{x^2}{4} \cos x$$

$$\text{i.e. } \frac{1}{D^2+1} x \cos x = \frac{x}{4} \cos x + \frac{x^2}{4} \sin x \quad \text{and} \quad \frac{1}{D^2+1} x \sin x = \frac{x}{4} \sin x - \frac{x^2}{4} \cos x$$

$\therefore$ The general solution is $y = y_c + y_p$

$$\therefore y = A \cos x + B \sin x + \frac{1}{4} (x \sin x - x^2 \cos x) \quad (\text{Ans.})$$

(ii) **Hints.** [In similar manner find $y_c = A \cos x + B \sin x$ and from the above

$$y_p = \frac{1}{D^2+1} x \cos x = \frac{x}{4} \cos x + \frac{x^2}{4} \sin x, \text{ hence G.S. solution is } y = A \cos x + B \sin x + \frac{1}{4} (x \cos x + x^2 \sin x).]$$

Example-2 Solve the differential equations:

$$(i) (D^2 - 1)y = x^2 \cos x$$

$$(ii) (D^2 - 1)y = x^2 \sin x$$

Solution (i) Given differential equation is $(D^2 - 1)y = x^2 \cos x \quad \dots\dots(1)$

Let $y = e^{mx}$ be a trial solution of the equation (1), then the auxiliary equation of the corresponding homogeneous equation of (1) is $m^2 - 1 = 0 \quad \therefore m = \pm 1$.

Therefore, the complementary function $y_c = Ae^{-x} + Be^x$.

Now, the particular integral of the differential equation (1) is

$$\begin{aligned} y_p &= \frac{1}{D^2-1} x^2 \cos x \\ &= \text{Real part of } \frac{1}{D^2-1} x^2 e^{ix} \\ &= \text{Real part of } e^{ix} \frac{1}{(D+i)^2-1} x^2 \\ &= \text{Real part of } e^{ix} \frac{1}{D^2+2Di-2} x^2 \\ &= \text{Real part of } e^{ix} \frac{1}{-2 \left\{ 1 - \left(Di + \frac{D^2}{2} \right) \right\}} x^2 \\ &= \text{Real part of } -\frac{e^{ix}}{2} \left\{ 1 - \left(Di + \frac{D^2}{2} \right) \right\}^{-1} x^2 \\ &= \text{R. p. of } -\frac{e^{ix}}{2} \left\{ 1 + \left(Di + \frac{D^2}{2} \right) + \left(Di + \frac{D^2}{2} \right)^2 + \dots \right\} x^2 \end{aligned} \quad \left| \begin{aligned} &= \text{Real part of } -\frac{e^{ix}}{2} \left\{ 1 + Di + \frac{D^2}{2} - D^2 - \dots \right\} x^2 \\ &= \text{Real part of } -\frac{e^{ix}}{2} (x^2 + 2xi + 1 - 2 - 0) \\ &= \text{Real part of } -\frac{1}{2} (\cos x + i \sin x) \{ (x^2 - 1) + 2xi \} \\ &= \text{Real part of} \\ &\quad -\frac{1}{2} \left[\{ (x^2 - 1) \cos x - 2x \sin x \} + i \{ (x^2 - 1) \sin x + 2x \cos x \} \right] \\ &= -\frac{1}{2} (x^2 - 1) \cos x + x \sin x \\ &\therefore \text{ The general solution is } y = y_c + y_p \\ &\therefore y = Ae^{-x} + Be^x - \frac{1}{2} (x^2 - 1) \cos x + x \sin x \quad (\text{Ans.}) \end{aligned} \right.$$

(ii) **[Hints.** Similarly find $y_c = Ae^{-x} + Be^x$ and from the above $y_p = \frac{1}{D^2+1} x^2 \sin x =$ Imaginary part of

$$\frac{1}{D^2+1} x^2 e^{ix} = -\frac{1}{2} (x^2 - 1) \sin x - x \cos x, \text{ hence G.S. solution is } y = Ae^{-x} + Be^x - \frac{1}{2} (x^2 - 1) \sin x - x \cos x]$$

Example-3 Solve the differential equations:

(i) $(D^2 - 2D + 1)y = x \sin x$

(ii) $(D^2 - 2D + 1)y = x \cos x$

Solution (i) Given differential equation is $(D^2 - 2D + 1)y = x \sin x$ (1)

Let $y = e^{mx}$ be a trial solution of the equation (1), then the auxiliary equation of the corresponding homogeneous equation of (1) is

$$m^2 - 2m + 1 = 0 \quad \therefore m = 1, 1$$

Therefore, the complementary function $y_c = (c_1 + c_2 x)e^x$.

Now, the particular integral of the differential equation (1) is

$$\begin{aligned} y_p &= \frac{1}{D^2 - 2D + 1} x \sin x \\ &= \text{Imaginary part of } \frac{1}{D^2 - 2D + 1} x e^{ix} \\ &= \text{Imaginary part of } e^{ix} \frac{1}{(D+i)^2 - 2(D+i) + 1} x \\ &= \text{Imaginary part of } e^{ix} \frac{1}{D^2 + 2Di + i^2 - 2D - 2i + 1} x \\ &= \text{Imaginary part of } e^{ix} \frac{1}{D^2 + 2Di - 2D - 2i} x \\ &= \text{Imaginary part of } \frac{ie^{ix}}{2} \frac{1}{1 - \left(2D + 2Di - \frac{iD^2}{2}\right)} x \\ &= \text{Imaginary part of } \frac{ie^{ix}}{2} \left[1 - \left(2D + 2Di - \frac{iD^2}{2}\right)\right]^{-1} x \\ &= \text{Imaginary part of } \frac{ie^{ix}}{2} \left[1 + \left(2D + 2Di - \frac{iD^2}{2}\right) + \dots\right] x \\ &= \text{Imaginary part of } \frac{ie^{ix}}{2} [x + (2 + 2i - 0) + 0] \\ &= \text{Imaginary part of } \frac{i(\cos x + i \sin x)}{2} (x + 2 + 2i) \\ &= \text{Imaginary part of } \frac{i}{2} [\{(x+2)\cos x - 2\sin x\} + i\{(x+2)\sin x + 2\cos x\}] \\ &= \text{Imaginary part of } \frac{1}{2} [i\{(x+2)\cos x - 2\sin x\} - \{(x+2)\sin x + 2\cos x\}] \\ &= \frac{1}{2} [\{(x+2)\cos x - 2\sin x\}] \end{aligned}$$

$\therefore$ The general solution is $y = y_c + y_p$

$$\therefore y = (c_1 + c_2 x)e^x - \sin x + \frac{1}{2}(x+2)\cos x \text{ (Ans.)}$$

(ii) [**Hints.** Similarly find $y_c = (c_1 + c_2x)e^x$ and from the above $y_p = \frac{1}{D^2 - 2D + 1} x \sin x$
 $= \text{Real part of } \frac{1}{D^2 - 2D + 1} x e^{ix} = -\frac{1}{2} \{(x+2)\sin x + 2\cos x\}$, hence G.S. solution is
 $y = (c_1 + c_2x)e^x - \cos x - \frac{1}{2}(x+2)\sin x]$

Example-4 Solve the differential equations:

(i) $(D^2 - 2D + 1)y = x e^x \sin x$

(ii) $(D^2 - 2D + 1)y = x e^x \cos x$

Solution (i) Given differential equation is $(D^2 - 2D + 1)y = x e^x \sin x$ (1)

Let $y = e^{mx}$ be a trial solution of the equation (1), then the auxiliary equation of the corresponding homogeneous equation of (1) is

$$m^2 - 2m + 1 = 0 \quad \therefore m = 1, 1$$

Therefore, the complementary function $y_c = (c_1 + c_2x)e^x$.

Now, the particular integral of the differential equation (1) is

$$\begin{aligned} y_p &= \frac{1}{D^2 - 2D + 1} x e^x \sin x \\ &= \frac{1}{(D-1)^2} x e^x \sin x \\ &= e^x \frac{1}{(D+1-1)^2} x \sin x \\ &= e^x \frac{1}{D^2} x \sin x \\ &= e^x \left[x \frac{1}{D^2} \sin x - \frac{2D}{(D^2)^2} \sin x \right] \\ &= e^x \left[x \frac{1}{-1^2} \sin x - \frac{2D}{(-1^2)^2} \sin x \right] \\ &= e^x [-x \sin x - 2 \cos x] \end{aligned}$$

$\therefore$ The general solution is $y = y_c + y_p$

$$\therefore y = (c_1 + c_2x)e^x - e^x [2 \cos x + x \sin x] \text{ (Ans.)}$$

(ii) [**Try yourself**]

Example-5 Solve the differential equations:

(i) $(D^2 - 4D + 4)y = 8x^2 e^{2x} \sin 2x$

(ii) $(D^2 - 4D + 4)y = 8x^2 e^{2x} \cos 2x$

Solution (i) The general solution is $y = y_c + y_p$

$$\therefore y = (c_1 + c_2x)e^{2x} + e^{2x} \{-2x^2 \sin 2x - 4x \cos 2x + 3 \sin 2x\} \text{ (Ans.)}$$

(ii) [**Try yourself**]